

Baseline Scales and Phenomenological Consistency in Modal Triplet Theory

Peter Nero

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Version 2

Abstract

We provide a typed scale and provenance ledger for Modal Triplet Theory (MTT). The internal vertical gap, complementary stabilization rate, coherent contraction rate, four-dimensional EFT cutoff, curvature scale, Planck scale, and renormalization scale are distinct objects. The internal gap is an internal mass or truncation scale only after the operator normalization and physical action are specified; it does not automatically damp high four-dimensional energy or satisfy particle, fifth-force, Lorentz-violation, gravitational-wave, general-relativistic, or cosmological bounds. We state the conditional Kaluza–Klein-type bridge under which internal eigenvalues become four-dimensional masses, retain the Foundation’s Schur–Feshbach truncation estimate, and define a formal consistency set built from sourced observable-response maps. A phenomenological consistency claim requires a nonempty common set or an exhibited parameter witness with theory errors and provenance. No such numerical witness is constructed here, so this paper makes no prediction and no assertion of present empirical viability. Its purpose is to prevent scale conflation and to specify exactly what downstream calculations must supply.

Version 2 Revision Note

Supersedes The first edition of *Baseline Scales and Phenomenological Consistency in Modal Triplet Theory*.

Reason The internal spectral gap, coherent damping, external EFT cutoff, curvature scale, Planck scale, and RG scale had been treated as if one number controlled them all.

Resolution Version 2 supplies separate typed scales, normalization and provenance requirements, a conditional internal-mode mass bridge, and a formal common consistency set.

Retained result Internal gaps still control truncation and complementary modes under the stated operator assumptions.

Remaining boundary No empirical witness or held-out prediction is claimed until observable maps, uncertainties, and a common parameter witness are supplied.

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1 Role and claim status

This paper is a ledger, not a phenomenological fit. It introduces no new physical dynamics and assigns no numerical values. It distinguishes:

Structural condition. A mathematical gate such as a spectral or semigroup bound.

Physical bridge. A theorem connecting an abstract operator to a term in a selected physical action.

Calibration. A parameter value inferred from measured input.

Consistency witness. One common parameter point satisfying declared constraints.

Held-out prediction. An observable not used in construction, calibration, scale setting, or branch selection.

The absence of an identified contradiction is not evidence that a common parameter point exists. Conversely, a freely adjustable parameter choice is a consistency witness, not a prediction.

2 Units and normalization

Every scale must be accompanied by units. In natural units, a Laplace-type operator has mass dimension two only after a physical length normalization has been selected. If an internal operator is initially dimensionless, write

$$A_{\text{int}}^{\text{phys}} = M_{\text{int}}^2 \hat{A}_{\text{int}}$$

with dimensionless $\hat{A}_{\text{int}}$ and a separately sourced conversion scale M_{int} . Quoting an eigenvalue without this normalization does not define a particle mass or energy cutoff.

Rates generated by the stabilization parameter τ have units of τ^{-1} . They are not physical decay widths or inverse seconds unless a bridge between τ and physical time t is supplied.

3 Structural scale registry

3.1 Internal spectral gap

Let

$$A_{\text{int}} \succeq \lambda_*^{\text{int}} Q$$

on the noncoherent vertical sector. The quantity λ_*^{int} is an internal spectral separation. With a mass-squared normalization, $m_*^{\text{int}} = \sqrt{\lambda_*^{\text{int}}}$ is the corresponding internal mode scale.

The internal gap supports a vertical resolvent estimate and, under the generator bridge of the Foundation, complementary stabilization. It does not by itself:

- define a four-dimensional momentum cutoff;
- damp modes merely because their base energy is high;
- remove every light field from a physical spectrum;
- determine a coupling to visible matter; or
- fix a Planck, cosmological, or RG scale.

3.2 Projector and mixing bounds

Record the Sobolev norms

$$C_P(s) = \|P\|_{H^s \rightarrow H^s}$$

and the typed mixing blocks L_{PQ}, L_{QP} . Under the domain hypotheses of the Foundation Schur–Feshbach theorem,

$$\|L_{PQ} L_{QQ}^{-1} L_{QP}\| \leq \|L_{PQ} L_{QQ}^{-1}\| \|L_{QP}\|.$$

Only after proving an inverse bound in terms of the internal gap or a stable semigroup may this be rewritten as an inverse-gap estimate. The norms, domains, and units of all three factors must be stated.

3.3 Complementary stabilization rate

The noncoherent linearized generator is controlled by

$$\|e^{\tau L_{QQ}}\| \leq M_Q e^{-\omega_Q \tau}.$$

Here M_Q captures nonnormal transient amplification and ω_Q is a stabilization rate. In the special bounded-perturbation model

$$L_{QQ} = -\kappa A_{\text{int}} + B_Q,$$

one may take $\omega_Q = \kappa \lambda_*^{\text{int}} - \|B_Q\| > 0$. This relation is conditional on the displayed generator and its invariant domain.

3.4 Coherent contraction scale

For a time-step map on an invariant basin,

$$\|T_\tau u - T_\tau v\| \leq q_{\text{coh}} \|u - v\|, \quad 0 \leq q_{\text{coh}} < 1.$$

The dimensionless factor q_{coh} , or a rate $\mu_{\text{coh}} = -\tau^{-1} \log q_{\text{coh}}$, controls basin-local convergence. It is independent of λ_*^{int} unless a realization proves a relation.

3.5 Leakage and curved-cluster margins

Curvature or parameter variation introduces at least two different scales: the spectral perturbation norm $\|\mathcal{R}\|$ and the off-diagonal leakage $\|Q\mathcal{R}P\|$. FP IV controls cluster persistence and the old-projector leakage floor under explicit smallness and damping margins. These quantities are not automatically physical spacetime-curvature corrections to Einstein's equations.

4 Physical scales requiring a completion

4.1 Four-dimensional EFT cutoff

Denote the cutoff of a selected four-dimensional effective action by Λ_{4D} . It is the scale at which that action loses controlled accuracy because of heavy thresholds, strong coupling, higher derivatives, nonlocality, unitarity limits, or omitted states. It is not definitionally equal to m_*^{int} .

At most, a proved physical reduction may yield a relation such as

$$\Lambda_{4D} \leq \min\{m_{\text{heavy}}, \Lambda_{\text{strong}}, \Lambda_{\text{derivative}}, \Lambda_{\text{nonlocal}}\}.$$

Which entry is limiting is model dependent.

Theorem 4.1 (Conditional internal-mass bridge). *Suppose a local physical scalar action on $Y_4 \times X_6$ contains the quadratic operator*

$$-\square_4 \otimes I + I \otimes A_{\text{int}} + m_0^2$$

with product measure, compatible domains, and an orthonormal internal eigenbasis $A_{\text{int}}\phi_k = \lambda_k\phi_k$. Then expansion $\Phi(x, z) = \sum_k \varphi_k(x)\phi_k(z)$ gives four-dimensional quadratic masses

$$m_k^2 = m_0^2 + \lambda_k.$$

Proof. Insert the eigenfunction expansion in the quadratic action and integrate over X_6 . Orthogonality diagonalizes the internal operator. $\square$

This familiar bridge is not supplied by abstract projection alone. Warping, mixing, boundary conditions, interactions, and normalization can change the mass matrix.

4.2 External high-energy damping

Suppression of large four-dimensional momentum requires an external base operator, form factor, dissipative physical kernel, or EFT truncation theorem. A vertical gap distinguishes internal eigenmodes but acts on every base momentum carried by a fixed internal mode. Therefore the statement “high four-dimensional energy is exponentially damped by the internal gap” is not available without an external damping construction.

4.3 Planck and curvature scales

The Planck scale M_{Pl} is the coefficient of the selected four-dimensional gravitational action. In a particular higher-dimensional Einstein–Hilbert compactification one may derive a volume relation such as

$$M_{\text{Pl}}^2 = M_{10}^8 V_6$$

up to conventions, warping, and corrections. The abstract MTT internal gap does not imply this formula or fix any factor in it.

Higher-curvature scales $\Lambda_{R^2}, \Lambda_{R^3}, \dots$ are coefficients in a physical derivative expansion. A schematic expression like $\varepsilon_{\text{warp}}/\lambda_*^{\text{int}}$ is dimensional analysis only until the physical action and matching calculation are supplied.

4.4 Renormalization scale

The RG scale μ_{RG} labels a family of renormalized descriptions. It is not stabilization time, the internal eigenvalue, or the EFT cutoff, though a matching calculation may compare them at a threshold.

5 Observable-response requirements

An experimental constraint applies to an observable response, not directly to an abstract spectral gap. The following rows illustrate the missing bridges.

5.1 New particles and fifth forces

A fifth-force test constrains both the physical mediator mass and its coupling to matter. A Yukawa correction has schematic form

$$V(r) = -\frac{Gm_1m_2}{r} [1 + \alpha_5 e^{-m_5 r}] .$$

Even if Theorem 4.1 identifies m_5 with an internal eigenvalue, the strength α_5 requires a coupling and matter-charge calculation. A large internal gap alone is not a fifth-force certificate.

5.2 Lorentz and equivalence-principle tests

Lorentz violation is constrained through coefficients in the physical principal symbol and interactions. Equivalence-principle violation is constrained through species-dependent couplings and trajectories. Neither is bounded by λ_*^{int} without an explicit response map.

5.3 Gravitational waves

Gravitational-wave observations constrain quantities such as propagation speed, damping, dispersion, polarization, and source generation in a selected gravitational model. These depend on the base principal symbol and physical action. The internal gap may enter after matching, but it is not itself a GW observable.

5.4 General relativity and cosmology

Tests of GR constrain post-Newtonian parameters, lensing, binary dynamics, wave generation, and other sourced observables. Cosmology constrains a background solution and perturbations, including expansion, growth, and primordial data. Conditions such as $\lambda_*^{\text{int}} \gg H_0^2$ are dimensionally meaningful only after normalization and are not sufficient to establish agreement with these data.

6 Formal consistency set

Let Θ be a declared parameter and branch space, and let $\Theta_{\text{struct}} \subseteq \Theta$ be the subset satisfying every structural gate. For each empirical row a , specify:

- a response map $F_a : \Theta_a \rightarrow \mathcal{O}_a$ derived from the same physical completion;
- a domain condition $\Theta_a \subseteq \Theta$;
- a theory-error set $E_a(\theta)$;
- an experimental acceptance set I_a ; and
- provenance showing which data selected θ , the branch, and the form of F_a .

Define

$$\mathcal{C} = \Theta_{\text{struct}} \cap \bigcap_a \{\theta \in \Theta_a : F_a(\theta) + E_a(\theta) \text{ intersects } I_a\}.$$

Proposition 6.1 (Consistency-witness criterion). *The declared model is jointly consistent with the listed rows exactly when $\mathcal{C}$ is nonempty. An exhibited $\theta_* \in \mathcal{C}$ is a consistency witness. It is a held-out prediction only for rows whose data did not enter the model, calibration, scale choice, branch choice, or uncertainty construction.*

This proposition is a bookkeeping identity, but it prevents a serious logical error: satisfying different bounds with different unlinked parameter choices does not exhibit one viable model.

7 Fine tuning, parameter count, and falsifiability

The statement “no fine tuning is required” needs a declared measure, prior, sensitivity, or naturalness criterion on Θ . Possible diagnostics include the local response Jacobian, condition numbers, or the fraction of a declared prior volume contained in $\mathcal{C}$. No such measure is selected here.

The ledger must classify every adjustable quantity as:

1. structurally derived;
2. independently measured input;
3. calibrated parameter;
4. nuisance/theory-error parameter; or

5. discrete branch or model choice.

Parameter reduction can be claimed only when source theorems reduce this list before empirical replay.

The framework is falsified for a specified completion if $\mathcal{C} = \emptyset$, a required operator/domain gate fails, or a held-out observable lies outside its acceptance set after declared errors.

8 Current status of the ledger

The corrected Foundation and Fixed Points papers provide several structural rows: internal gap, semigroup damping, contraction, leakage, truncation, and admissibility conditions. The signature paper provides the principal-symbol compatibility ledger. They do not provide one selected physical action and a common set of response maps covering particle, fifth-force, Lorentz, GR, GW, and cosmological observables.

Accordingly, this paper does not establish that current empirical bounds are simultaneously satisfied. It also identifies no contradiction. The phenomenological status is *not evaluated by this ledger* until the physical maps and a common consistency witness are supplied.

9 Downstream citation rules

Downstream papers may cite this work for:

- the distinction among internal, control, EFT, curvature, Planck, and RG scales;
- the conditional internal-mass bridge;
- the consistency-set protocol; and
- the required parameter and provenance classifications.

They may not cite it as proof that MTT satisfies experimental constraints, has no fine tuning, predicts a scale, suppresses all UV physics, or reproduces GR or cosmology.

10 Conclusion

MTT contains several useful but logically distinct scales. The internal gap can control vertical truncation and, in a specified product action, contribute to four-dimensional masses. It is not a universal cutoff or empirical shield. Phenomenological consistency begins only after one physical completion emits observable-response maps and one common parameter point survives all sourced constraints. This ledger now states that task without prejudging its outcome.

References

- [1] P. Nero, *Modal Triplet Theory: Foundations*, revised v7, 2026.
- [2] P. Nero, *Fixed Points I–VI*, corrected theorem spine, 2026.
- [3] P. Nero, *Lorentzian Base Compatibility and Signature Stability in the MTT Fixed-Point Realization*, revised v2, 2026.